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Systematic Review Or Meta-Analysis

Risk of dementia in bipolar disorder and the interplay of lithium: a systematic review and meta-analyses

Velosa J, Delgado A, Finger E, Berk M, Kapczinski F, de Azevedo Cardoso T. Risk of dementia in bipolar disorder and the interplay of lithium: a systematic review and meta-analyses.

Objectives: To assess whether bipolar disorder (BD) increases the rate of dementia and whether lithium is related to a lower risk of dementia in BD.
Methods: A total of 10 studies (6859 BD; 487 966 controls) were included in the meta-analysis to test whether BD is a risk factor for dementia. In addition, five studies (6483 lithium; 43 496 non-lithium) were included in the meta-analysis about the potential protective effect of lithium in BD.
Results: BD increases the risk of dementia (odds ratio (OR): 2.96 [95% CI: 2.09–4.18], $P < 0.001$), and treatment with lithium decreases the risk of dementia in BD (OR: 0.51 [95% CI: 0.36–0.72], $P < 0.0001$). In addition, secondary findings from our systematic review showed that the risk of progression to dementia is higher in BD than in major depressive disorder (MDD). Moreover, the number of mood episodes predicted the development of dementia in BD.
Conclusion: Individuals with BD are at higher risk of dementia than both the general population or those with MDD. Lithium appears to reduce the risk of developing dementia in BD.

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Key words: aging; Alzheimer's; bipolar disorder; dementia; depression; lithium; mania; neuroprogression; prevention; treatment

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Summations

- The diagnosis of bipolar disorder increases the risk to develop dementia almost three-fold.
- Treatment with lithium decreased the risk for dementia by 49% in individuals with bipolar disorder compared to controls/non-lithium treated people with BD.
- Dementia can be seen as an endpoint of bipolar disorder staging model.

Limitations

- There is a possibility that publication bias may have potentially inflated our results.
- The majority of the evidence comes from registry-based studies that have several limitations such as data collection, reliability of diagnosis, selection bias, and confounders.
- Additional studies are needed to understand the pathophysiology of BD that leads to cognitive deterioration, brain abnormalities, and to the development of dementia.

Introduction

Bipolar disorder (BD) is an episodic illness with a heterogeneous clinical course. The lifetime prevalence of BD ranges from 0.6% to 2.4% worldwide (1, 2). Many studies have shown that during the course of BD, cognitive impairment is an important feature in both acute and euthymic phases (3). A systematic review showed that although the clinical course of mood disorders is heterogenous, both BD and MDD demonstrate evidence of clinical progression. In this sense, a higher number of mood episodes seems to be related to (i) increased risk of recurrence, (ii) longer episodes, (iii) increased severity of episodes, (iv) decreased threshold for subsequent episodes, and (v) increased risk for dementia (4). However, conflicting data about aging and cognitive decline are found in the literature, and despite positive findings, studies did not demonstrate faster cognitive deterioration in BD compared to controls (5, 6).

One important consideration is whether cognitive impairment in BD may progress to dementia. To our knowledge, there is only one published meta-analysis showing that a BD diagnosis is a risk factor for dementia (7). In addition, despite some evidence about the neuroprotective effects of lithium (in terms of protecting brain structure, relating to increases in hippocampal volume and preventing white matter changes; and cognition, relating to improvements in verbal memory (8–11)), no meta-analysis has been performed to assess whether treatment with lithium decreases the risk of developing dementia in patients with BD. Thus, we aimed to: (i) update the literature examining BD as a risk factor for developing dementia and (ii) verify whether lithium protects against dementia in patients with BD.

Methods

The Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) guidelines were followed for the present review.

Search strategy

A literature search was conducted on July 8, 2019, according to the PRISMA statement, with no year or language restrictions, using the following databases: Pubmed, PsycInfo, and Scopus. We searched for a combination of the following search items (bipolar disorder OR bipolar disorders OR mania OR hypomania OR manic OR hypomanic OR manic-depressive disorder OR manic depressive disorder OR bipolar affective disorder) AND (dementia OR Alzheimer OR frontotemporal dementia). The search yielded 5080 articles: (Pubmed = 1556, PsycInfo = 1452, and Scopus = 2472), with 3926 remaining after duplicate removal.

To determine whether an article was relevant to our study, we used the following inclusion criteria: The study should (i) present original data and (ii) require a diagnosis of bipolar disorder preceding a diagnosis of dementia. The exclusion criteria were as follows: (i) reviews and meta-analyses, (ii) case reports, and (iii) studies where dementia was considered as comorbidity and not as an outcome.

The studies were assessed by two blinded raters (AD and JV) who determined whether studies met inclusion criteria. The two raters assessed manuscripts independently, and divergences were resolved in a meeting with another researcher (TC). First, the raters screened articles by title and abstract and after by full-text. All articles not fulfilling the search criteria were excluded. The details of the search strategy are presented in Fig. 1.

Data extraction

Two researchers (AD and JV) were involved in the data extraction process. We extracted: authorship, year of publication, country where the study took place, study aims, study design, assessments, and main results.

Quality assessment

Each manuscript included was independently assessed by two blind researchers (AD and JV) using the Newcastle-Ottawa Quality Assessment Scale (NOQAS) (12). Disagreements were resolved by consensus in a meeting with another researcher (TC).

Statistical analysis

Random effects meta-analyses were performed using the RevMan 5.3. We conducted two analyses: (i) To analyze the studies that investigated BD as a risk factor for dementia, we used the number of cases diagnosed with dementia in both bipolar disorder and non-bipolar disorder groups to calculate the odds ratio (OR) and (ii) we analyzed the studies that assessed whether treatment with lithium was associated with a reduced risk of dementia in patients with BD. For this aim, we used the number of cases diagnosed with dementia in lithium users and non-lithium users groups to calculate the odds ratio (OR). Significance was set as $P < 0.05$. Cochrane's Q test was performed to

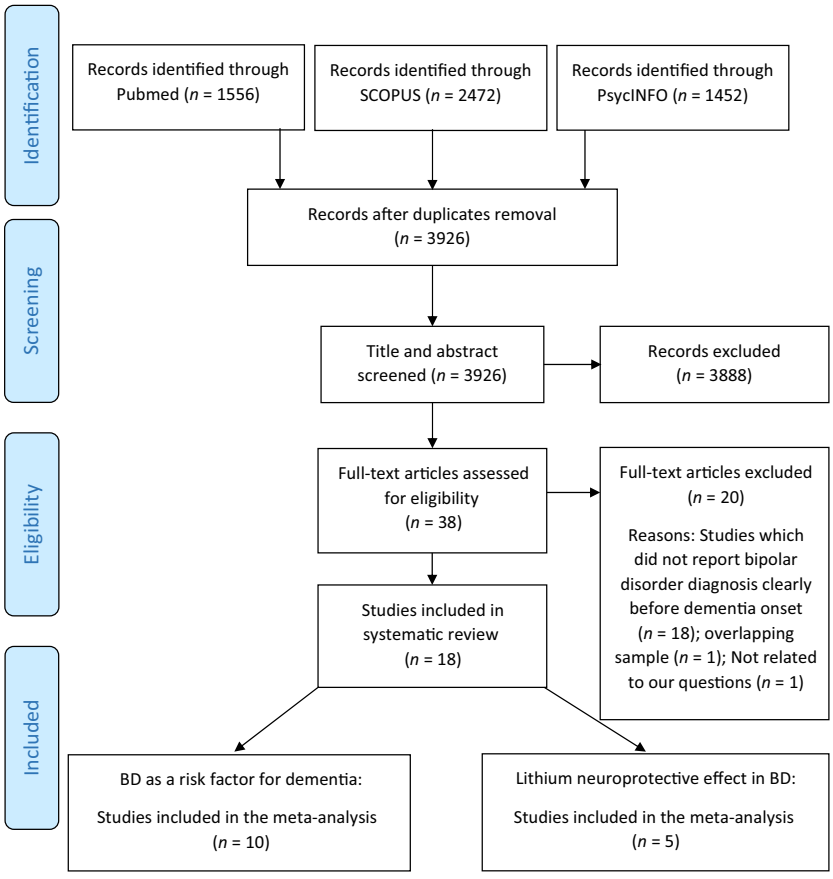

Fig. 1. Flow diagram showing the search, article selection, and extraction process.. [Colour figure can be viewed at wileyonlinelibrary.com]

assess for statistical heterogeneity, and the Higgins I^2 statistic was used to determine the extent of variation between sample estimates with values ranging from 0 to 100%. Studies were not included in the meta-analyses if the population overlapped with another included study population or if it was not possible to calculate the OR. If the information was not reported in the paper, we contacted the authors to request access to the data. We contacted four authors, and three of them answered our questions.

Results

The literature search yielded 5480 studies. Of these, 1554 were duplicates, and 3888 studies were excluded as the titles and abstracts were not relevant to the research topic, leaving 38 potentially eligible studies for which the full text was reviewed. After this stage, 20 studies did not meet the inclusion criteria. A total of 18 studies met all criteria to be included in the systematic review (Fig. 1). In addition, we searched the references of the included studies and found no additional studies to include.

From the 18 included studies, 10 (four cohort, six case–control studies) were included in the meta-analysis to assess whether BD is a risk factor for dementia, and 5 (four cohort, one case–control studies) were included in the meta-analysis to assess whether lithium reduces the risk of dementia in BD. Finally, four cohort studies presenting preliminary evidence of differences between BD and MDD regarding the risk for developing dementia were included, and one study assessed which clinical characteristics contribute to risk for developing dementia in BD was also included, both as secondary outcomes of our systematic review.

Increased risk for dementia in bipolar disorder

Characteristics of included studies. Table 1 shows an overview of the included studies. Among the 11 studies, publication dates ranged from 1998 to 2018. Three studies were conducted in Taiwan, three in Australia, two in Denmark, one in the United Kingdom, one in Finland, and one in Brazil. The total sample size ranged from 209 to 190 041. All studies assessed the impact of BD on the development of dementia. Five studies had a longitudinal cohort design, and six studies were case–control. For the cohort studies, the time of follow-up assessments ranged from 3 to 17 years. Regarding the assessment of BD, the majority of the studies used ICD-9 (n = 7), and the diagnosis

Table 1. Characteristics of the studies included for the assessment whether bipolar disorder as a risk factor for dementia

Author, Year, Country	Aim	Study design	Assessments	Main results	Quality	Is BD a risk factor for D?
Cooper, 1998, United Kingdom	To test whether the risk for D is increased by a history of earlier psychiatric illness.	Case–Control/ Registry-based	ICD-8, ICD-9	*n* total = 1118; BD = 6 (0.5%) BD to *D* = 6 (100%) Controls to *D* = 553 (49.7%)	8/9	Yes
Kessing, 2003, Denmark	To assess whether MDD or BD have an increased risk of developing D compared to other chronic illnesses	Cohort/Registry-base	ICD-8	$n_{osteoarthritis}$ total = 83 387; BD = 2007 (2.4%) BD to *D* = 38 (1.9%) (OR 3.38, 95% CI 2.39–4.79) Controls to *D* = 895 (1.1%) $n_{diabetes}$ total = 71 156; BD = 2007 (2.8%) BD to *D* = 38 (1.9%) (OR 2.28, 95% CI 1.62–3.20) Controls to *D* = 691 (1%)	7/9	Yes
Laursen, 2011, Denmark	To investigates the impact of severe chronic diseases on excess mortality in individuals with psychotic disorders	Cohort/Registry-base	ICD-8, ICD-10	Dementia had the highest IRR (IRR: 11.52, 95% CI 6.76–19.65) in individuals with BD	8/9	Yes†
Wu, 2013, Taiwan	To investigate whether BD patients were at an increased risk for developing D.	Case–Control/ Registry-based	ICD-9	*n* total = 64 804; BD = 229 (0.4%) BD to *D* = 114 (49.8%) (OR 4.32, 95% CI 3.21–5.82) Controls to *D* = 9190 (14.2%)	8/9	Yes
Aprahamian, 2014, Brazil	To evaluate the performance of the CDT, in patients with BD with and without probable AD.	Case–Control	DSM-IV	*n* total = 209; BD = 106 (50.7%) BD to *D* = 27 (25.5%) Controls to *D* = 33 (32%)	5/9	No‡
Zilkens, 2014, Australia	To assess the association of history of psychiatric disorders with D.	Case–Control/ Registry-base	ICD-10	*n* total = 27 136; BD = 68 (0.3%) BD to D = 59 (86.8%) (OR 4.71, 95% CI 2.29–9.65) Controls to *D* = 13 509 (49.9%)	8/9	Yes
Chen, 2015, Taiwan	To assess the risk of dementia among BD or MDD.	Cohort/Registry-base	ICD-9	*n* total = 2408; BD = 345 (14.3%) BD to *D* = 92 (26.7%) (HR 5.58, 95% CI 4.26–7.32) MDD to *D* = 322 (HR 3.02, 95% CI 2.46–3.70) Controls to *D* = 133 (6.2%)	7/9	Yes
Taipale, 2015, Finland	To investigate the prevalence of BZD and BZDR use, and associated factors among individuals with and without AD	Case–Control/ Registry-based	ICD-8, ICD-9, ICD-10	*n* total = 49 951; BD = 225 (0.4%) BD to *D* = 133 (59.1%) Controls to *D* = 24 833 (49.9%)	8/9	Yes‡
Almeida, 2016, Australia	To investigate the 13-year risk of dementia and death in older adults with BD.	Cohort/Registry-base	ICD-8, ICD-9, ICD-10	*n* total = 37 768; BD = 256 (0.7%) BD to *D* = 65 (25.4%) (HR = 2.30, 95% CI 1.80–2.94). Controls to *D* = 4860 (13%)	9/9	Yes
Cheng, 2017, Taiwan	To assess the risk of AD associated with use of lithium.	Case–Control/ Registry-based	ICD-9	*n* total = 190 041; BD = 3537 (1.9%) BD to *D* = 2074 (58.6%) (OR 2.34, 95% CI 2.18–2.51) Controls to *D* = 61 274 (32.8%)	8/9	Yes
Almeida, 2018, Australia	To examine associations between age of onset of BD and clinical comorbidities as well as D and mortality.	Cohort/Registry-base	ICD-8, ICD-9, ICD-10	*n* total = 38 003; BD = 80 (0.2%) BD to *D* = 6 (7.5%) (OR 17.74, 95% CI 7.52–41.84) Controls to *D* = 383 (1%)	9/9	Yes

D, dementia; BD, bipolar disorder; IRR, incidence rate ratios; CDT, clock-drawing test; AD, Alzheimer's disease; BZD, benzodiazepine; BZDR, benzodiazepine and related drugs; MDD, major depressive disorder; N/A, non-applicable; DSM, diagnostic statistic manual of mental disorders criteria; ICD, international statistical classification of diseases and related health problems criteria.

†Not included in the meta-analysis (no available data).

‡Based on OR value from our meta-analysis

of dementia was mainly performed using ICD-9 (n = 7).

Increased risk for dementia in bipolar disorder: evidence from cohort studies. The systematic review included five cohort studies assessing whether BD is a risk factor for dementia. All studies showed that BD presents with an increased risk for developing dementia (13–17). Kessing et al. (13) included all inpatients with a main diagnosis of BD, major depression, osteoarthritis, or diabetes from January 1, 1977, to December 31, 1993, with no history of dementia before selection. In this study, BD patients had a higher risk of being diagnosed with dementia (HR 3.38, 95% CI 2.39–4.79 vs. controls with osteoarthritis; HR 2.28, 95% CI 1.62–3.20 vs. controls with diabetes). This study suggests that the cognitive impairment in mood

disorders patients is not explained by a chronic disorder but by the mood disorder itself (13). The impact of 19 severe chronic diseases in patients with BD and Schizophrenia was reported by Laursen et al. (14) in a population-based cohort study. The highest incidence rate ratios (IRR) for dementia were observed in both disorders (Schizophrenia: 5.84, 95% CI: 3.96–9.25; and BD: 11.52, 95% CI: 6.76–19.65) in comparison with the individuals without psychiatric contact (14).

Additionally, Chen et al. (15) included individuals older than 55 years with MDD or BD and without dementia in a 10-year period. Diagnosis of any type of dementia was identified during the 3-year follow-up or until death. The findings showed that BD had a higher risk of developing dementia when compared to controls (HR 5.58, 95% CI 4.26–7.32) (15). Moreover, Almeida et al. (16) included men aged 65–85 years, without dementia until April 2, 1996. Past diagnosis of BD was retrieved from another database that has been recorded prior to April 2, 1996. Follow-up started from this date until the outcome appeared (dementia or death) or until June 30, 2009 (database endpoint). Their findings showed that BD had a higher adjusted hazard of dementia (HR 2.30, 95% CI 1.80–2.94). Their data showed that elderly men with past diagnosis of BD are at higher risk of dementia which is greater for those with either a long-standing or recent history of BD. Thus, it appears that changes in mood later in life may represent, in at least some cases, an early clinical expression of neurodegenerative processes that will ultimately lead to the diagnosis of dementia (16). Finally, Almeida et al. (17) included men aged 65–85 during 1996–1998 from a specific Australian population. Then, they retrieved information about the diagnosis of BD from another database and started a follow-up until December 31, 2013. The results showed that regardless of the age of BD onset, these individuals had an increased risk for dementia (Age < 50 - HR 3.46, 95% CI 2.41–4.95; Age ≥ 50–HR 2.26, 95% CI 1.76–2.90). Incident dementia was higher when onset of BD illness was before 50 years and after 60 years, even though this effect was not significant (17).

Increased risk for dementia in bipolar disorder: evidence from case–control studies. Six case–control studies were included in the systematic review. In five of them, BD patients had higher risk of developing dementia (18–22), while one study did not find this association (23). Cooper et al. (18) included all residents above 60 years from a database including patients with dementia from May 1993 to April 1995. They then matched the ones with a previous history of psychiatric disorder with other patients with dementia without history of psychiatric illness (controls). Their results suggest that history of psychiatric disorders, especially a major mental disorder like BD, appears to be a risk for dementia. They also found that this association is not linked to a specific subtype of dementia, like Alzheimer's disease (AD) (18). Furthermore, Wu et al. (19) selected patients with 45 years or above who were diagnosed with dementia between January 2000 and December 2009. Then, they searched for the presence of a previous BD diagnostic. The findings demonstrated that the diagnosis of dementia is higher in BD patients as compared to controls (OR 4.07; 95% CI 3.08–5.37), and these patients had greater risk for both presenile and senile dementia (19).

Moreover, Zilkens et al. (20) included all persons with incident dementia between 2000 and 2009 aged 65 to 84 years, from four databases. Then, they assessed the exposure of medical risk factors such as BD, which had to be present 10 years before dementia onset. Their results showed that BD had a higher risk for dementia (OR 4.71, 95% CI 2.29–9.65) (20). In addition, Taipale et al. (21) included all patients with AD diagnosis alive on December 31, 2005 (n = 28093) and matched a comparison individual without that diagnosis. They searched for psychiatry history, including BD, before the onset of AD. Although the aim of the study was not to determine whether BD is a risk factor for dementia, their data show a higher risk of being diagnosed with dementia in subjects with BD (21). Finally, Cheng et al. (22) included individuals with 65 years old or older who had an AD diagnosis, from a national database between January 2007 and December 2009. For each subject, two controls without dementia were matched. Their findings suggest that BD had a higher risk of developing dementia (OR 2.34, 95% CI 2.18–2.51) (22).

In contrast, Aprahamian et al. (23) included older adults whom they divided into two groups: one without dementia and a second with mild AD. Their results showed no association between BD and dementia. However, their findings demonstrated that BD patients commonly had cognitive impairments, even in euthymia (23).

Increased risk for dementia in bipolar disorder: results from the meta-analysis. Our meta-analysis included 10 out of the 11 studies described above, including 6859 BD individuals and 487 966 non-BD, and our results confirmed that the diagnosis of BD is associated with an almost three-fold increase in the risk of developing dementia (OR: 2.96 [95%

CI: 2.09 - 4.18], $P < 0.001$) (Fig. 2). The funnel plot for this association was asymmetric indicating potential publication bias (Figure S1).

Effects of lithium

Characteristics of included studies. Table 2 shows an overview of the included studies. Among the six studies, publication dates ranged from 2007 to 2017. Two studies were conducted in the United States, one in Taiwan, one in Denmark, one in Switzerland, and one in Brazil. Total sample size ranged from 114 to 41 251. All studies assessed the use of lithium and the development of dementia in BD individuals. Five studies had a longitudinal cohort design, and one study was case–control.

Neuroprotective effect of lithium: results from the extant literature. Six studies assessed the neuroprotective effect of lithium in patients with BD (22, 24–28). Four of these studies found that lithium seems to reduce the risk of dementia in BD patients.

Angst et al. (24) included 406 patients with major mood disorders and did not find a reduced risk of dementia with any medication (including lithium). However, when examining only patients with BD, long-term lithium and clozapine use were associated with better functioning in demented patients (24). In addition, Nunes et al. (25) in a study including 114 elderly patients with BD found that recent use of lithium was associated with a reduced prevalence of AD. Their results indicate that lithium seems to reduce the risk of dementia in patients with BD and raises the question of whether lithium may have neuroprotective effects that protect against the development of dementia in BD (25).

Moreover, Kessing et al. (26) included 4856 patients with BD doing different treatments (lithium, anticonvulsants, antidepressants, and antipsychotics) using a registry-based study. The findings suggest that only continued treatment with lithium was related to a lower rate of dementia (26). Finally, Gerhard et al. (27) included 41 931 patients and demonstrated that continued use of lithium in elderly BD patients was associated with a lower incidence of dementia (hazard ratio (HR) = 0.77, 95% CI 0.60–0.99).

Gildengers et al. (28) found no differences between lithium or divalproex users vs. those not exposed regarding the development of dementia in an underpowered study including 33 patients with BD. Also, Cheng et al. (22) in a study with 63 347 patients with AD found that in overall population, there was an increased risk of AD with lithium use (as well as other mood stabilizers, valproic acid, and carbamazepine). However, no excess risk was discernible among people with BD (aOR = 1.36; 95% CI, 0.89–2.09). No association was seen between any use of lithium and AD risk, nor for associations between higher levels of use (cumulative dose, cumulative period of use, or mean daily dose) and AD risk (22).

Neuroprotective/neurotrophic effect of lithium: results from the meta-analysis. We included five out of six studies described above in our meta-analysis. The four studies included 6483 lithium users and 43 496 non-lithium users. Our results showed that lithium treatment decreased the risk for dementia by 49% in BD individuals (OR: 0.51 [95% CI: 0.36–0.72], $P < 0.0001$) (Fig. 3). The funnel plot for this association was symmetric indicating no publication bias (Figure S2).

Secondary findings from the systematic review

Differences between bipolar disorder vs. major depressive disorder and rates of dementia. The systematic

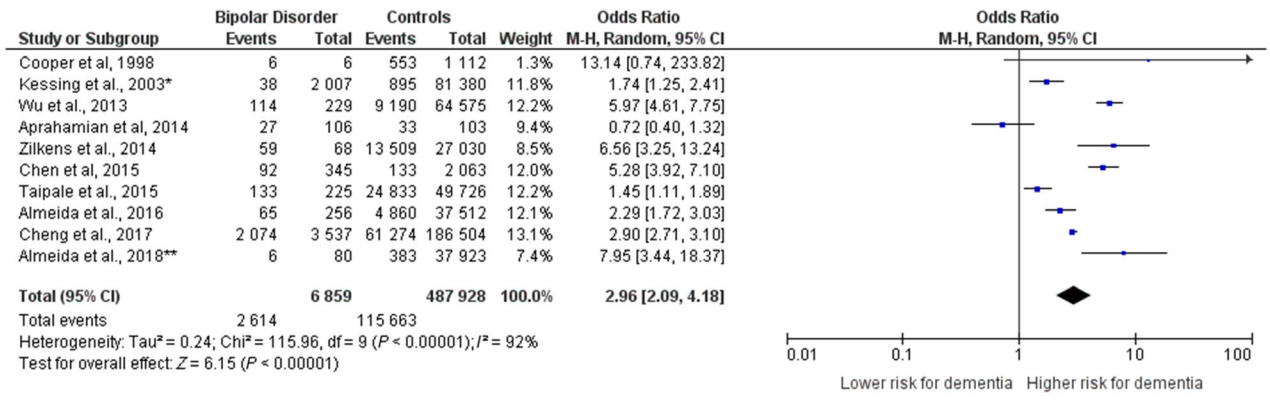

Fig. 2. Forest plot of the meta-analysis of BD as risk factor for dementia. [Colour figure can be viewed at wileyonlinelibrary.com]

Table 2. Characteristics of the studies included for the assessment whether the neuroprotective/neurotrophic effect of lithium

Author, Year, Country	Aim	Study design	Assessments	Main results	Quality	Lithium protect against D in BD?
Angst, 2007, Switzerland	To investigate whether long-term medication with lithium, clozapine or antidepressants prevent or attenuate dementia in BD and MDD	Cohort	ICD-9	*n* total = 220; Lithium = 76 (34.5%) Lithium to *D* = 14 (18.4%) Controls to *D* = 26 (18.1%)	9/9	No‡
Nunes, 2007, Brazil	To assess whether the continuous lithium use as a protector against AD	Cohort	DSM-IV	*n* total = 114; Lithium = 66 (57.9%) Lithium to *D* = 3 (4.5%) (OR 0.079, 95 CI 0.020–0.321) Controls to *D* = 16 (33.3%)	6/9	Yes
Gildengers, 2009, USA	To investigate whether older adults with BD would present with more cognitive dysfunction and a more rapid cognitive decline	Cohort	DSM-IV	*n* total = 33; Lithium = 9 (27.7%) No significant difference between BD patients exposed to lithium or divalproex vs. those not exposed regarding the development of dementia	7/9	No†
Kessing, 2010, Denmark	To investigate whether treatment with lithium in patients with BD decrease the rate of dementia	Cohort/Registry-based	ICD-8, ICD-10	*n* total = 4856; Lithium = 2449 (50.4%) Lithium to *D* = 84 (3.4%) Controls to *D* = 132 (5.5%)	8/9	Yes
Gerhard, 2015, USA	To examine the association of lithium and dementia risk in BD	Cohort/Registry-based	ICD-9	*n* total = 41 251; Lithium = 3766 (9.1%) Lithium to *D* = 64 (1.7%) (HR 0.77, 95 CI 0.60–0.99) Controls to *D* = 1377 (3.7%)	8/9	Yes
Cheng, 2017, Taiwan	To assess the risk of AD associated with use of lithium.	Case–Control/ Registry-based	ICD-9	*n* total = 3.537; Lithium = 125 (3.5%) Lithium to *D* = 32 (25.6%) Controls to *D* = 1431 (41.9%)	8/9	Yes‡

D, dementia; BD, bipolar disorder; MDD, major depressive disorder; AD, Alzheimer's disease; DSM, diagnostic statistic manual of mental disorders criteria; ICD, international statistical classification of diseases and related health problems criteria.
†Not included in the meta-analysis (no available data).
‡Based on OR value from our meta-analysis.

review included four studies assessing the differences between BD and MDD in terms of risk to develop dementia. Overall, the findings showed that both MDD and BD are associated with an increased risk for developing dementia (13, 29, 30). Additionally, the risk seems to be greater for BD as compared to MDD (15, 29, 30).

Kessing et al. (29) in a case-registry study, including patients with MDD (n = 3363), BD (n = 518), schizophrenia (n = 1025), and neurotic disorder (n = 8946), estimated the rate of developing dementia during 21 years of follow-up. The findings showed that BD patients had a higher risk of developing dementia, followed by MDD, schizophrenia, and finally neurotics (29). Also, Kessing et al. (30), in a study including 22 788 patients with a diagnosis of affective disorders, showed that the ones with one manic episode had an increased risk of dementia (HR 1.46 (95% CI 1.01 to 2.13)) when compared with the ones with one depressive episode (30). In a sample of 2291 patients aged 55 years with MDD or BD, Chen et al. (15) reported that BD patients had an 87% increased risk of developing dementia when compared to MDD patients during the study follow-up (HR 1.87, 95% CI 1.48–2.37). To evaluate the risk of receiving a dementia diagnosis, Kessing et al. (13) recruited a sample of patients with mania

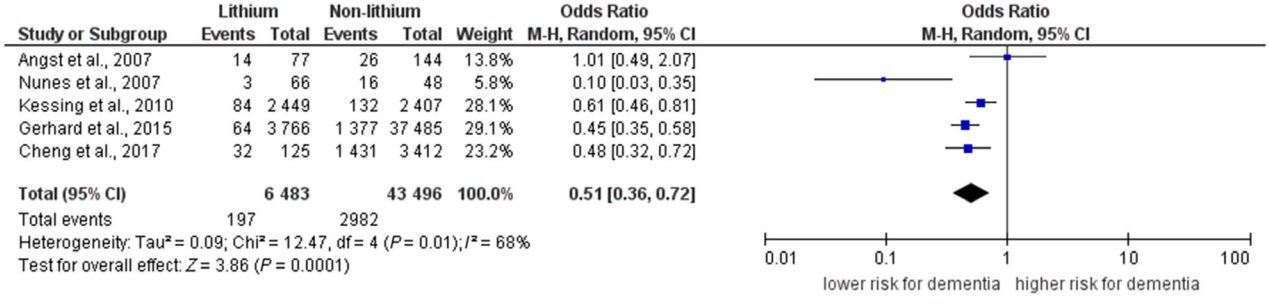

Fig. 3. Forest plot of the meta-analysis of neuroprotective/neurotrophic effect of lithium. [Colour figure can be viewed at wileyonline library.com]

(n = 2007), depression (n = 11 741), osteoarthritis (81 380), and diabetes (69 149) with a first-ever discharge from hospital between 1977 and 1993. The results showed that patients diagnosed with MDD at the first discharge had significantly higher risk of being diagnosed with dementia during long-term follow-up than patients with mania at first discharge (X^2 14.5, P = 0.0001) (13).

Clinical characteristics associated with the risk of dementia in bipolar disorder. In a study including 22 788 patients with the diagnosis of affective disorders, Kessing et al. (30) showed that the number of episodes foretold the rate of dementia: Every new episode leading to admission increased the rate by 13% (95% CI: 9% to 16%) for patients with MDD and by 6% (95% CI: 3–10) for patients with BD.

Quality of assessment of the included studies

For the BD progression to dementia meta-analysis, overall, the mean Newcastle-Ottawa Quality Assessment Scale score was 7.16. We identified eight good-quality studies and eleven poor-quality studies (Table S1). Most of the poor-quality studies failed in comparability and/or adequate follow-up.

Regarding lithium use and the risk for dementia in BD meta-analysis, overall, the mean NOQAS score was 7.67. We identified five good-quality studies and one poor-quality studies (Table S2). The poor-quality study failed in terms of adequate follow-up.

Discussion

This meta-analysis showed that BD is a risk factor for developing dementia. We also found that lithium is associated with a reduced risk of dementia in BD patients. In addition, secondary findings from the systematic review showed that the risk of progression to dementia is higher in BD than in MDD. Finally, the number of episodes in both MDD and BD patients predicts the progression to dementia.

Our findings are in-line with two prior meta-analyses showing that BD is a risk factor for dementia (7, 31, 32) and also with a recent large study (published after our literature search was performed) showing that BD is a predictor of Alzheimer's disease (33). Owing to our wider search strategy, we were able to include four older studies (not included in the most updated previous meta-analysis (7)) and two studies that were conducted after the publication of the most updated previous meta-analysis (7). It is important to highlight that BD and dementia present some similarities in the clinical manifestation (34–36) (agitation, psychotic, mood, and cognitive symptoms), regarding the structural brain neuroimaging (34, 37) (enlarged lateral ventricles and white matter hyperintensities using magnetic resonance imaging), and a recent study showed that both disorders presented a shared genetic basis (38). Those data together indicate a potential shared pathophysiology, which can, in turn, contribute for the misdiagnosis between these two disorders. Despite the above similarities, a recent study showed that a set of clinical, neuropsychological, and blood biomarker assessments were able to discriminate the two conditions, such as overall functioning, memory, and oxidative stress (39). As an attempt to minimize the bias of misdiagnosis in our systematic review and meta-analysis, we selected only studies of incident dementia in patients previously diagnosed as BD. By excluding the cases of patients with dementia who reported comorbid manic symptoms at baseline, we attempted to access the progression of the bipolar disease into dementia.

Our findings can be interpreted in light of the theory of neuroprogression (32, 40). Neuroprogression is marked by progressive cognitive (41) and functioning impairment (42), as well as altered brain scans and biomarkers of underlying processes (inflammation, oxidative stress, apoptosis, and neurogenesis) that are evident in some cases of bipolar disorder (43, 44). Recent evidence suggests that structural brain abnormalities in gray and white matter similar to those observed in age-related neurodegenerative diseases such as dementia can also be observed in BD and may be linked to accelerated brain aging (45, 46). Staging is a concept related to neuroprogression which has been introduced to explain progressive decline over the course of disease. A recent study showed that there is an association between later stages of BD and increasing levels of neurocognitive and social dysfunction (47). We could argue, based on our study and other meta-analyses, that the risk for dementia could be seen as a potential endpoint of this neuroprogressive course (7, 31). Related to this topic, it is important to assess whether cognitive impairment in BD may progress to dementia and whether those people with BD and cognitive complaints are the ones at risk of dementia. We did not found data to evaluate this topic but future studies should consider this as a possibility.

To the best of our knowledge, this is the first meta-analysis evaluating the risk of dementia associated with lithium use in BD patients. Conflicting evidence about lithium and dementia is found in

the literature. Dunn et al. (48) found that patients treated with lithium had an increased risk of being diagnosed with dementia. However, the study had some limitations such as small sample size of patients treated with lithium and the lack of psychiatric history. Also, Cheng et al. (22) found that lithium seems to increase the risk of AD in the general population. Nonetheless, this association was not observed when patients with BD were analyzed. This is important because lithium is mainly used in BD and, as discussed before, BD seems to be associated with a higher risk for dementia (22). On the other hand, Kessing et al. (49) found that only long-term use of lithium reduced the rate of dementia in a design taking account of confounding by indication. Long-term continued lithium treatment has been found associated with reduction of the rate of dementia to the same level as that for the general population (49). The association between the number of prescriptions for lithium and dementia was unique and different from the association between the number of prescriptions for anticonvulsants and dementia (49). All findings were replicated in subanalyses with AD as the outcome. Also, Kessing et al. (4) found that in general population, long-term increased lithium exposure in drinking water was associated with a lower incidence of dementia. Our meta-analysis showed that lithium is associated with reduced development of dementia in BD. This effect could be because of lithium's neuroprotective and neurotrophic properties. Differential mechanisms have been proposed for the neuroprotective effects associated with lithium: (i) Lithium inhibits glycogen synthase kinase-3 (GSK3) (50, 51); (ii) regulates calcium levels related with N-methyl-D-aspartate (NMDA) receptors; (iii) increases brain-derived neurotrophic factor (BDNF) (52); (iv) induces the repressor element 1-silencing transcription factor (REST) (53); and (v) reduces apoptotic markers and enhances neurogenesis and mitochondrial dysfunctions (54–56). These effects of lithium overlap with the known risk pathways for dementia—for a review of the neurobiology of lithium in dementia (57). Pilot studies of lithium in mild cognitive impairment showed that lithium may prevent further deterioration (58). Unfortunately, most lithium studies did not address the question of whether lithium users are healthier than non-users, raising the question whether there are other variables such as diabetes, vascular, thyroid, or kidney disease increasing the risk of dementia.

Multiple reviews have shown that MDD patients have a higher risk for dementia (31, 59, 60). Our data suggest that BD patients may have an even higher risk to develop dementia as compared to MDD. This is in-line with other meta-analysis showing that the risk for dementia is higher in BD than in MDD (BD 2.36 vs. MDD 1.65 to 2) (7, 59, 60). There are many possible explanations for these results: (i) different underlying pathophysiology despite symptom overlap, (ii) number of episodes—people with BD could have a higher number of episodes, which increases the risk for dementia, (iii) (hypo)manic episodes could have a stronger impact on risk for developing dementia as compared to depressive episodes, and (iv) time spent in an episode—BD could have a longer exposition to an episode, which can also increase the risk for dementia. Kessing et al. (30) reported that the number of episodes significantly predict the rate of dementia in both MDD and BD. The number of manic episodes seems to be the clinical marker most robustly associated with neuroprogression in BD (40). Clinical studies also describe an earlier onset, recurrent episodes, substance use disorder, medical comorbidities, and worse health-related behaviors as the risk factors (61, 62). In addition, BD is associated with inflammatory dysregulation (higher concentration of C reactive protein (CRP), IL-2 and IL-6 receptors, and tumor necrosis factor receptor 1 (TNF1)), which may also contribute to the development of dementia and neuroprogression in bipolar disorder (63–67).

Our findings should be interpreted in light of some limitations. As our meta-analysis included only published studies, it could be at risk of publication bias. Also, in order to perform the meta-analyses, we utilized non-adjusted results (absolute number of patients who developed dementia) which may have introduced some bias (healthcare utilization, comorbid medical and psychiatric disorders, and other psychotropic medication) in our results. Most of the studies included were registry-based which have several study-based limitations (data collection, diagnostic reliability, selection bias, and lack of information about confounders). Also, it is important to mention that usually the first manifestation of BD is at the early adulthood (68), while the diagnosis of dementia is more common after 65 years old (68). In this sense, despite the studies matched the sample (BD vs. non-BD) by age, our findings might be somewhat age-related. Despite these limitations, to the best of our knowledge, this is the first meta-analysis showing the potential effect of lithium preventing dementia in BD.

In conclusion, even though further studies are required to understand the pathophysiology of BD-related cognitive deterioration and the development of dementia, our meta-analysis showed

that BD patients are at higher risk for dementia when compared to the general population and patients with MDD. We also showed that lithium is associated with reduced risk of dementia in BD. This raises the question as to whether lithium may be a preventative agent in BD against the prospective development of dementia. Data from the present meta-analysis will set the stage to future randomized controlled trials to test the ability of lithium to prevent incident dementia among patients with BD.

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Declarations of interest

The authors have no conflict of interest in relation to this work.

Data availability statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Figure S1. Funnel plot for the studies included in the meta-analysis assessing whether bipolar disorder is a risk factor for dementia.

Figure S2. Funnel plot for the studies included in the meta-analysis assessing whether lithium is related to a lower risk of dementia in bipolar disorder.

Table S1. Risk of bias for cohort and case-control studies as assessed by The Newcastle-Ottawa Scale: Bipolar disorder as a risk factor for dementia.

Table S2. Risk of bias for cohort and case-control studies as assessed by The Newcastle-Ottawa Scale: Neuroprotective/neurotrophic effect of lithium